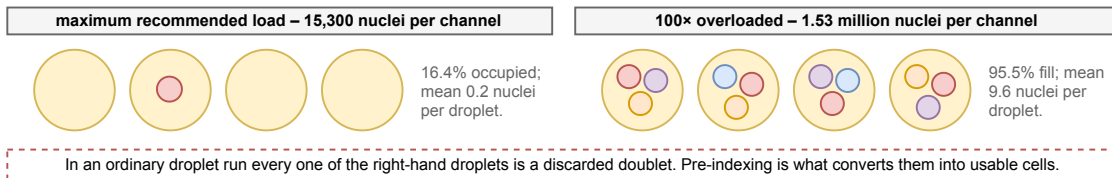


scifi-RNA-seq: pre-indexing the cells and droplet overloading stops being a failure mode

Combinatorial fluidic indexing of Datlinger et al. 2021. Poisson loading limits a droplet run only while each droplet must hold exactly one cell. Barcoding the cells first removes that requirement, so the same Chromium channel can be loaded a hundredfold harder. Colors and the 5'-to-3' convention follow Figs. 2 and 4.

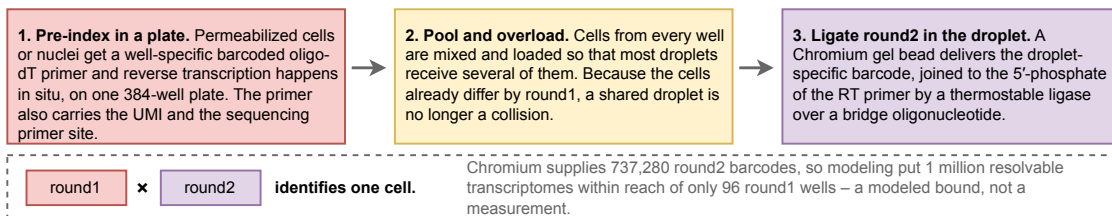
a What overloading a Chromium channel actually does, counted under a microscope

Both regimes ran without clogging the device. The droplets are drawn schematically; the counts are the measured means. In the separate yield run, 383,000 loaded nuclei returned 151,788 transcriptomes from one channel.

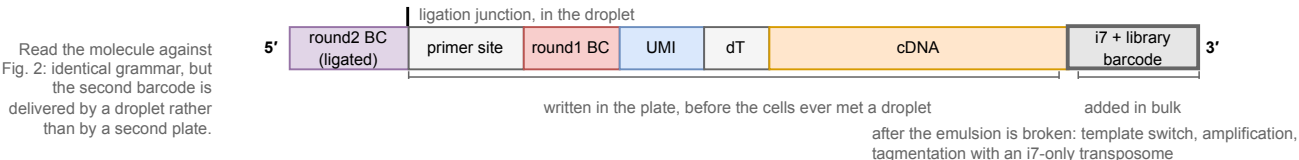


b Two rounds, two places: the plate writes round1 into the cell, the droplet ligates round2 on top

Round1 is shared by every cell from one well, round2 by every cell in one droplet. Neither alone identifies a cell.



c The molecule, in the notation of Figs. 2 and 4: round2 prepends to round1 by the same splint ligation split-pool uses



scifi-RNA-seq is the split-pool trick and the droplet trick composed, and the composition is what removes the Poisson tax. Split-pool writes a cell's address over successive plate rounds and never isolates anything (Fig. 2); droplet buys one container per cell and pays Poisson for it, keeping occupancy low (Fig. 4). Pre-indexing supplies the first half of the address in a plate, which leaves the droplet responsible only for the second half, so several cells per droplet becomes the operating point rather than the failure. The cost consequence: the per-channel reagent charge that dominates the droplet budget divides across 7.6x more cells, so 137 channels become 18 (Sec. 5.5).

